Loss of Markovianity in Gaussian AR(1) Diffusion

Precision fill-in, decay, and spectral interpretation

Abstract

A Gaussian AR(1) chain has a dense covariance but a tridiagonal precision. The tridiagonal pattern is the matrix form of Markov locality: conditionally on its neighbors, each frame is independent of the rest. After Ornstein–Uhlenbeck corruption,

$$\Sigma_t = e^{-2t}\Sigma_0 + (1 - e^{-2t})I, \qquad S(x,t) = \nabla_x \log p_t(x) = -\Sigma_t^{-1}(x - \mu_t),$$

so the full score structure is controlled by the noisy precision $Q_t := \Sigma_t^{-1}$. This note derives Q_0 from first principles, shows explicitly that any $t > 0$ produces nonzero long-range entries in Q_t, and explains the mechanism in two complementary ways. First, a Neumann expansion shows that diffusion fills the precision band one diagonal at a time. Second, a resolvent formula shows that Q_t is a diagonal matrix minus the inverse of a tridiagonal matrix, hence dense with exponential off-diagonal decay. Numerical experiments for small and large dimension confirm the lifecycle

$$\text{tridiagonal } Q_0 \longrightarrow \text{dense } Q_t \longrightarrow I \quad (t \to \infty).$$

1 Setup

Let $N := K + 1$ and consider the Gaussian AR(1) recursion

$$a_{k+1} = \alpha a_k + \eta_k, \qquad k = 0, 1, \ldots, K - 1, \tag{1}$$

where $a_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and the innovations $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$ are independent of each other and of a_0. We collect the trajectory into the vector

$$a := (a_0, a_1, \ldots, a_K)^\mathsf{T} \in \mathbb{R}^N.$$

The mean is immediate from the recursion:

$$\mu_k := \mathbb{E}[a_k] = \alpha^k \mu_0, \qquad \mu_a := (\mu_0, \alpha\mu_0, \ldots, \alpha^K \mu_0)^\mathsf{T}.$$

For most structural statements the mean is irrelevant, so we work with the centered vector

$$c := a - \mu_a.$$

2 The clean Gaussian law and its tridiagonal precision

2.1 Exact factorization of the quadratic form

Define the innovation coordinates

$$\xi_0 := c_0, \qquad \xi_k := c_k - \alpha c_{k-1}, \quad k = 1, \ldots, K.$$

By construction,

$$\xi_0 = a_0 - \mu_0, \qquad \xi_k = \eta_{k-1} \quad (k \geq 1),$$

so the vector $\xi = (\xi_0, \ldots, \xi_K)^\mathsf{T}$ is Gaussian with independent coordinates and covariance

$$D := \operatorname{diag}(\sigma_0^2, \sigma_\eta^2, \ldots, \sigma_\eta^2).$$

Introduce the lower bidiagonal matrix

$$L = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\alpha & 1 & 0 & \cdots & 0 \\ 0 & -\alpha & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\alpha & 1 \end{pmatrix}.$$

Then $\xi = Lc$. Since $\xi \sim \mathcal{N}(0, D)$,

$$p_0(a) \propto \exp\left(-\frac{1}{2}\xi^\mathsf{T} D^{-1}\xi\right) = \exp\left(-\frac{1}{2}c^\mathsf{T} L^\mathsf{T} D^{-1} Lc\right).$$

Therefore the clean precision is

$$Q_0 := \Sigma_0^{-1} = L^\mathsf{T} D^{-1} L. \tag{2}$$

2.2 Expanding the entries of Q_0

We now multiply out (2) entry by entry.

Proposition 2.1 (Explicit clean precision)**.** *The clean precision matrix is tridiagonal:*

$$Q_0 = \begin{pmatrix} \sigma_0^{-2} + \alpha^2\sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} & 0 & \cdots & 0 \\ -\alpha\sigma_\eta^{-2} & (1+\alpha^2)\sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} & \ddots & \vdots \\ 0 & -\alpha\sigma_\eta^{-2} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & (1+\alpha^2)\sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} \\ 0 & \cdots & 0 & -\alpha\sigma_\eta^{-2} & \sigma_\eta^{-2} \end{pmatrix}. \tag{3}$$

Proof. Because L has only one diagonal and one subdiagonal, $L^\mathsf{T} D^{-1} L$ can only have one diagonal and one super/subdiagonal. We compute them explicitly.

For the $(1, 1)$ entry only the first column of L matters. It has nonzero entries 1 in row 1 and $-\alpha$ in row 2, hence

$$(Q_0)_{11} = \sigma_0^{-2} + \alpha^2\sigma_\eta^{-2}.$$

For an interior diagonal entry $2 \leq i \leq N - 1$, the ith column of L has entries 1 in row i and $-\alpha$ in row $i + 1$, both weighted by σ_η^{-2}, so

$$(Q_0)_{ii} = \sigma_\eta^{-2} + \alpha^2\sigma_\eta^{-2} = (1 + \alpha^2)\sigma_\eta^{-2}.$$

For the last diagonal entry only row N contributes, giving

$$(Q_0)_{NN} = \sigma_\eta^{-2}.$$

Finally, for each neighboring pair $(i, i + 1)$, the only common nonzero row in the ith and $(i + 1)$th columns of L is row $i + 1$, where the entries are $-\alpha$ and 1. Therefore

$$(Q_0)_{i,i+1} = (Q_0)_{i+1,i} = -\alpha\sigma_\eta^{-2}.$$

All other entries vanish because the corresponding columns of L have disjoint support. This is exactly (3). □

2.3 Why this tridiagonal pattern is the Markov property

For a Gaussian vector $X \sim \mathcal{N}(\mu, \Sigma)$ with precision $Q = \Sigma^{-1}$, the conditional law of X_i given all other coordinates is obtained by isolating all terms in the quadratic form that contain x_i:

$$-\frac{1}{2}(x-\mu)^{\mathsf{T}} Q(x-\mu) = -\frac{1}{2} Q_{ii}(x_i-\mu_i)^2 - (x_i-\mu_i)\sum_{j\neq i} Q_{ij}(x_j-\mu_j) + \text{terms independent of } x_i.$$

Completing the square gives

$$X_i \,\big|\, X_{-i} \sim \mathcal{N}\left(\mu_i - Q_{ii}^{-1}\sum_{j\neq i} Q_{ij}(X_j-\mu_j),\ Q_{ii}^{-1}\right). \tag{4}$$

Hence $Q_{ij} = 0$ means that X_j does not enter the conditional mean of X_i once the other coordinates are known. For Gaussian vectors this is exactly conditional independence.

Applying (4) to Q_0 in (3), every interior coordinate a_k depends conditionally only on a_{k-1} and a_{k+1}. In the stationary zero-mean case $\sigma_0^2 = \sigma_\eta^2/(1-\alpha^2)$ this becomes especially transparent:

$$\mathbb{E}[a_k \mid a_{-k}] = \frac{\alpha}{1+\alpha^2}(a_{k-1}+a_{k+1}), \qquad \operatorname{Var}(a_k \mid a_{-k}) = \frac{\sigma_\eta^2}{1+\alpha^2}, \qquad 1 \le k \le K-1. \tag{5}$$

So the sparsity of Q_0 is not a secondary fact. It is the algebraic form of Markov locality.

3 OU diffusion: covariance and score

Let $z \sim \mathcal{N}(0, I_N)$ be independent of a, and define the noisy observation

$$x = e^{-t}a + \sqrt{\Delta_t}\, z, \qquad \Delta_t := 1 - e^{-2t}, \qquad \beta_t := e^{-2t}. \tag{6}$$

Then

$$\mu_t := \mathbb{E}[x] = e^{-t}\mu_a, \tag{7}$$

and, because a and z are independent,

$$\Sigma_t := \operatorname{Cov}(x) = \beta_t \Sigma_0 + \Delta_t I. \tag{8}$$

Since x is Gaussian, its log-density is quadratic and its score is affine:

$$S(x,t) = \nabla_x \log p_t(x) = -Q_t(x-\mu_t), \qquad Q_t := \Sigma_t^{-1}. \tag{9}$$

The entire question is therefore: *what happens to the matrix Q_t?*

4 The first nonzero long-range entry: a three-frame calculation

Take the stationary zero-mean case and set $N = 3$. Then

$$\Sigma_0 = \sigma_\infty^2 \begin{pmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{pmatrix}, \qquad \sigma_\infty^2 = \frac{\sigma_\eta^2}{1-\alpha^2}.$$

Write

$$u := \Delta_t + \beta_t \sigma_\infty^2, \qquad v := \beta_t \sigma_\infty^2 \alpha, \qquad w := \beta_t \sigma_\infty^2 \alpha^2.$$

Then

$$\Sigma_t = \begin{pmatrix} u & v & w \\ v & u & v \\ w & v & u \end{pmatrix}.$$

The $(1,3)$ entry of the inverse is the corresponding cofactor divided by $\det \Sigma_t$:

$$(Q_t)_{13} = \frac{v^2 - uw}{\det \Sigma_t}. \tag{10}$$

Now compute the numerator explicitly:

$$v^2 - uw = \beta_t^2 \sigma_\infty^4 \alpha^2 - (\Delta_t + \beta_t \sigma_\infty^2)(\beta_t \sigma_\infty^2 \alpha^2) = -\beta_t \Delta_t \sigma_\infty^2 \alpha^2.$$

Therefore

$$(Q_t)_{13} = -\frac{\beta_t \Delta_t \sigma_\infty^2 \alpha^2}{\det \Sigma_t}. \tag{11}$$

For every $t > 0$ and every $\alpha \neq 0$, the denominator is positive because Σ_t is positive definite, while the numerator is strictly negative. Hence

$$(Q_t)_{13} \neq 0 \qquad (t > 0).$$

At $t = 0$ this entry is exactly zero; the instant diffusion begins, the zero is filled.

5 Mechanism of fill-in: two exact matrix identities

5.1 Small-time band generation

Starting from (8), factor out β_t:

$$\Sigma_t = \beta_t\big(\Sigma_0 + c_t I\big), \qquad c_t := e^{2t} - 1 = \frac{\Delta_t}{\beta_t}.$$

Since $\Sigma_0 = Q_0^{-1}$,

$$\Sigma_t = \beta_t Q_0^{-1}(I + c_t Q_0),$$

so

$$Q_t = e^{2t}(I + c_t Q_0)^{-1} Q_0. \tag{12}$$

If $c_t \|Q_0\| < 1$, then the Neumann series converges:

$$Q_t = e^{2t} \sum_{m=0}^{\infty} (-c_t)^m Q_0^{m+1}. \tag{13}$$

This formula explains the fill-in mechanically. Because Q_0 is tridiagonal, Q_0^{m+1} has bandwidth $m+1$. Therefore:

- $m = 0$ contributes the original tridiagonal matrix Q_0;
- $m = 1$ adds the second off-diagonal band;
- $m = 2$ adds the third off-diagonal band;
- and so on.

Diffusion does not destroy locality in one jump. It adds longer and longer conditional paths order by order.

5.2 Exact resolvent form

There is a second identity, valid for all $t > 0$ and more useful globally. Write (8) as

$$\Sigma_t = \Delta_t(I + \gamma_t \Sigma_0), \qquad \gamma_t := \frac{\beta_t}{\Delta_t}.$$

Then

$$Q_t = \frac{1}{\Delta_t}(I + \gamma_t \Sigma_0)^{-1}. \tag{14}$$

Since $I + \gamma_t \Sigma_0 = Q_0^{-1}(Q_0 + \gamma_t I)$,

$$(I + \gamma_t \Sigma_0)^{-1} = (Q_0 + \gamma_t I)^{-1} Q_0,$$

and therefore

$$Q_t = \frac{1}{\Delta_t}\Big[I - \gamma_t (Q_0 + \gamma_t I)^{-1}\Big]. \tag{15}$$

So every off-diagonal entry of Q_t comes from the inverse of the tridiagonal matrix $B_t := Q_0 + \gamma_t I$.

Proposition 5.1 (Every off-diagonal entry becomes nonzero)**.** *Fix $t > 0$ and assume $\alpha \neq 0$. Then every off-diagonal entry of Q_t is nonzero.*

Proof. Write $b := \alpha \sigma_\eta^{-2} > 0$ and index matrices from 1 to N. The matrix B_t is tridiagonal with positive diagonal entries $d_1, \ldots, d_N$ and constant off-diagonals $-b$. Define the leading and trailing determinant recursions

$$\theta_0 = 1, \qquad \theta_1 = d_1, \qquad \theta_k = d_k \theta_{k-1} - b^2 \theta_{k-2} \quad (k \geq 2),$$

$$\phi_{N+1} = 1, \qquad \phi_N = d_N, \qquad \phi_k = d_k \phi_{k+1} - b^2 \phi_{k+2} \quad (k \leq N-1).$$

Because B_t is symmetric positive definite, all principal minors are positive, hence all θ_k and ϕ_k are positive. The explicit inverse formula for a tridiagonal matrix gives, for $i < j$,

$$(B_t^{-1})_{ij} = b^{j-i} \frac{\theta_{i-1}\phi_{j+1}}{\theta_N} > 0.$$

Using (15),

$$(Q_t)_{ij} = -\frac{\gamma_t}{\Delta_t}(B_t^{-1})_{ij} < 0 \qquad (i \neq j).$$

So no off-diagonal entry can vanish. □

6 Quantifying the loss of locality

6.1 Finite-N metrics

For any matrix $M \in \mathbb{R}^{N \times N}$ and any radius $r \geq 1$, define the off-diagonal band share

$$R_r(M) := \frac{\sum_{1 \leq |i-j| \leq r} |M_{ij}|}{\sum_{i \neq j} |M_{ij}|}. \tag{16}$$

We use it on $M = Q_t$ and write $R_r(t) := R_r(Q_t)$. Then:

- $R_1(0) = 1$ because Q_0 is exactly tridiagonal;

- $R_1(t) < 1$ for every $t > 0$ by Proposition 5.1;
- as $t \to \infty$, all off-diagonal entries vanish and $Q_t \to I$, so the total coupling strength goes to zero.

Thus the lifecycle is not monotone. Diffusion first creates longer-range conditional couplings, then eventually erases all couplings as isotropic noise dominates.

A second useful diagnostic is the distance profile

$$C_d(t) := \frac{1}{\#\{(i,j) : |i-j| = d\}} \sum_{|i-j|=d} |(Q_t)_{ij}|, \qquad d = 1, 2, \dots, N-1. \tag{17}$$

This measures how strongly the precision couples coordinates at graph distance d.

6.2 Stationary infinite-chain formula

For a clean stationary chain the covariance is Toeplitz,

$$(\Sigma_0)_{ij} = \sigma_\infty^2 \alpha^{|i-j|}, \qquad \sigma_\infty^2 = \frac{\sigma_\eta^2}{1-\alpha^2},$$

and the interior precision symbol is

$$q_0(\theta) = \frac{1 + \alpha^2 - 2\alpha\cos\theta}{\sigma_\eta^2}.$$

After diffusion,

$$q_t(\theta) = \frac{1}{\beta_t \sigma_\eta^2/(1+\alpha^2 - 2\alpha\cos\theta) + \Delta_t} = \frac{1+\alpha^2-2\alpha\cos\theta}{\beta_t\sigma_\eta^2 + \Delta_t(1+\alpha^2-2\alpha\cos\theta)}. \tag{18}$$

At $t = 0$, this is a first-degree trigonometric polynomial in $\cos\theta$, so its Fourier coefficients vanish beyond distance 1: that is exactly the tridiagonal precision. For every $t > 0$, the denominator depends on $\cos\theta$, so q_t is a rational function and its Fourier series contains all harmonics. That is the spectral version of fill-in.

The same statement can be written directly in physical space. Let

$$a_t := 1 + \alpha^2 + \gamma_t\sigma_\eta^2, \qquad \rho_t := \frac{a_t - \sqrt{a_t^2 - 4\alpha^2}}{2\alpha} \qquad (0 < \rho_t < 1,\ 0 < \alpha < 1).$$

Then the inverse of the infinite tridiagonal resolvent has Green's function

$$(Q_0 + \gamma_t I)^{-1}_{ij} = \frac{\sigma_\eta^2}{\sqrt{a_t^2 - 4\alpha^2}}\, \rho_t^{|i-j|},$$

and (15) gives, for $i \neq j$,

$$(Q_t)_{ij} = -A_t \rho_t^{|i-j|}, \qquad A_t := \frac{\gamma_t \sigma_\eta^2}{\Delta_t\sqrt{a_t^2 - 4\alpha^2}}. \tag{19}$$

So diffusion does *not* produce arbitrary dense structure. It produces a dense matrix with controlled exponential decay.

In this idealized translation-invariant setting, the off-diagonal band share has the closed form

$$R_r^{(\infty)}(t) = \frac{\sum_{d=1}^r \rho_t^d}{\sum_{d=1}^\infty \rho_t^d} = 1 - \rho_t^r. \tag{20}$$

Larger ρ_t means weaker locality.

7 Spectral interpretation: which cancellation is broken?

For finite N, diagonalize the clean covariance as

$$\Sigma_0 = U\Lambda U^\top, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N), \qquad U^\top U = I.$$

Then

$$Q_t = U\ \mathrm{diag}\left(\frac{1}{\beta_t\lambda_1 + \Delta_t}, \ldots, \frac{1}{\beta_t\lambda_N + \Delta_t}\right) U^\top. \tag{21}$$

At $t = 0$ the weights are $1/\lambda_m$. Their particular values cause exact cancellations among the dense rank-one matrices $u_m u_m^\top$, and the result is the tridiagonal matrix Q_0. Once diffusion changes the weights to $(\beta_t\lambda_m + \Delta_t)^{-1}$, those exact cancellations no longer hold. The matrix becomes full.

This also explains the late-time limit. As $t \to \infty$, all weights tend to 1, so (21) tends to $UIU^\top = I$. The endpoint is not "long-range Markov"; it is *no coupling at all.* Diffusion first breaks local conditional structure, then washes out structure entirely.

8 Numerical experiments

All figures were generated by the reproducible script `code/generate_figures.py`. We used the stationary choice $\alpha = 0.88$, $\sigma_\eta^2 = 1 - \alpha^2$, hence $\sigma_\infty^2 = 1$.

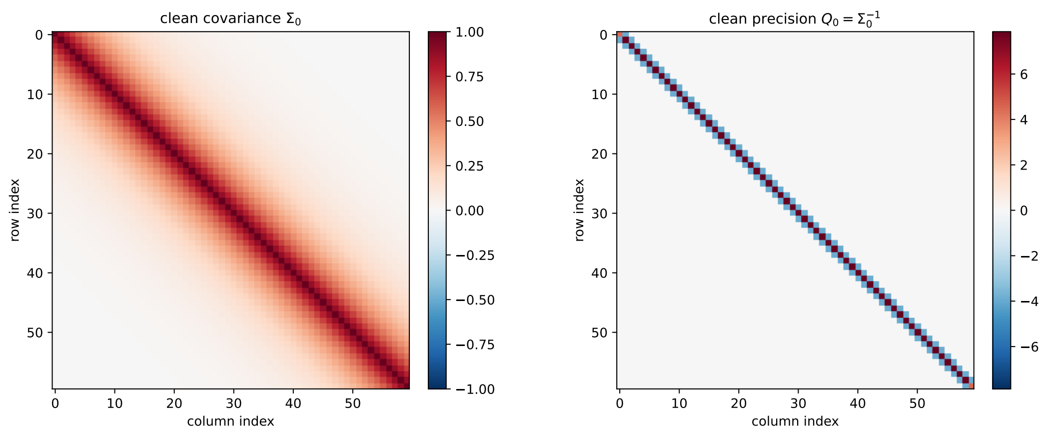

Figure 1: Clean covariance versus clean precision for $N = 60$. The covariance is dense because marginal correlations propagate through the recursion. The precision is tridiagonal because conditional dependence is local.

The numerical results support the analytic picture exactly:

1. The clean matrix Q_0 is sparse because the Gaussian AR(1) chain is conditionally local.
2. The first positive diffusion time already creates nonzero long-range entries.
3. These long-range entries are not arbitrary: they decay rapidly with distance, consistent with (19).
4. At late times the whole off-diagonal part collapses to zero and Q_t approaches the identity.

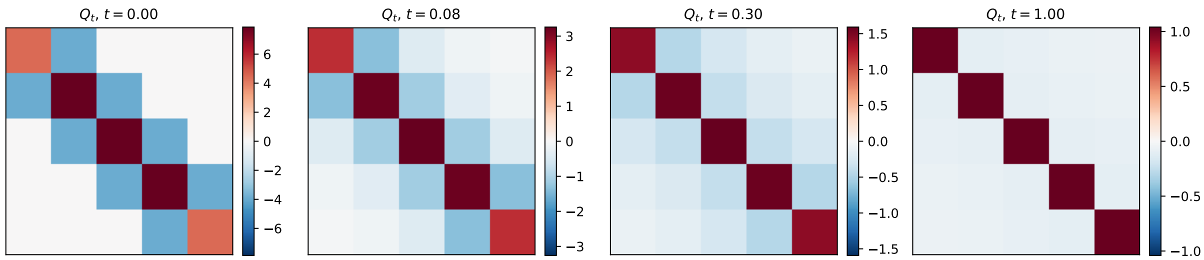

Figure 2: Small-N heatmaps of Q_t ($N = 5$). At $t = 0$ the matrix is exactly tridiagonal. For $t > 0$ the zeros fill in immediately. The small matrix makes the first long-range entries visually obvious.

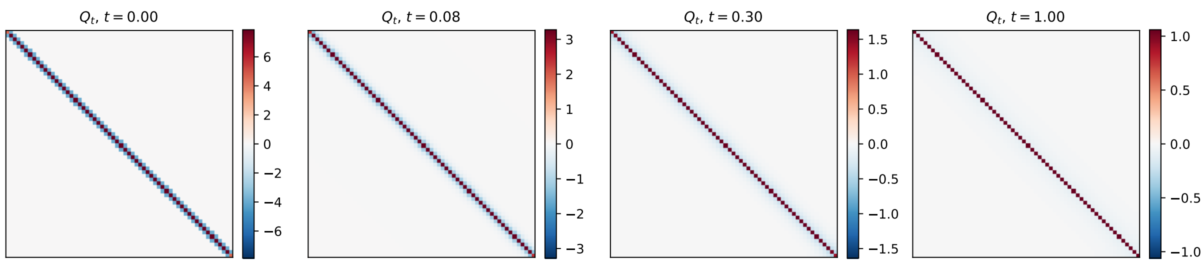

Figure 3: Large-N heatmaps of Q_t ($N = 60$). The fill-in persists at scale: diffusion creates a full matrix whose entries are strongest near the diagonal and then decay away from it.

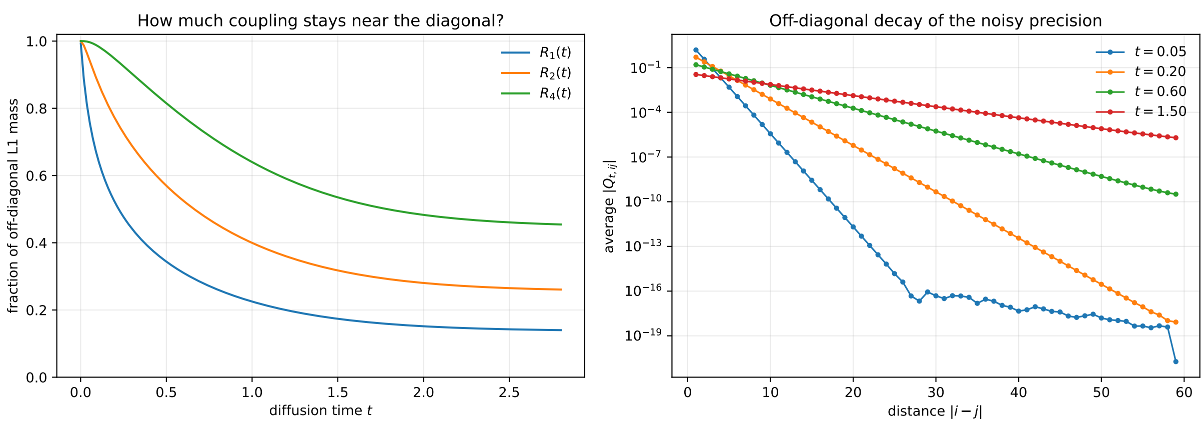

Figure 4: Left: nearest-band mass fractions $R_r(t)$ defined in (16). As soon as $t > 0$, some coupling leaves the nearest-neighbor band. Right: the averaged off-diagonal profile $C_d(t)$ defined in (17); the near-linear behavior on a semilog axis is the numerical signature of exponential decay.

9 Takeaway

There are three distinct objects in play:

$$\text{Markovianity} \quad \leftrightarrow \quad \text{sparse clean precision } Q_0,$$

$$\text{Gaussianity} \quad \leftrightarrow \quad \text{affine score } S(x,t) = -Q_t(x - \mu_t),$$

$$\text{diffusion} \quad \leftrightarrow \quad \text{change of inverse from } Q_0 \text{ to } Q_t = (\beta_t \Sigma_0 + \Delta_t I)^{-1}.$$

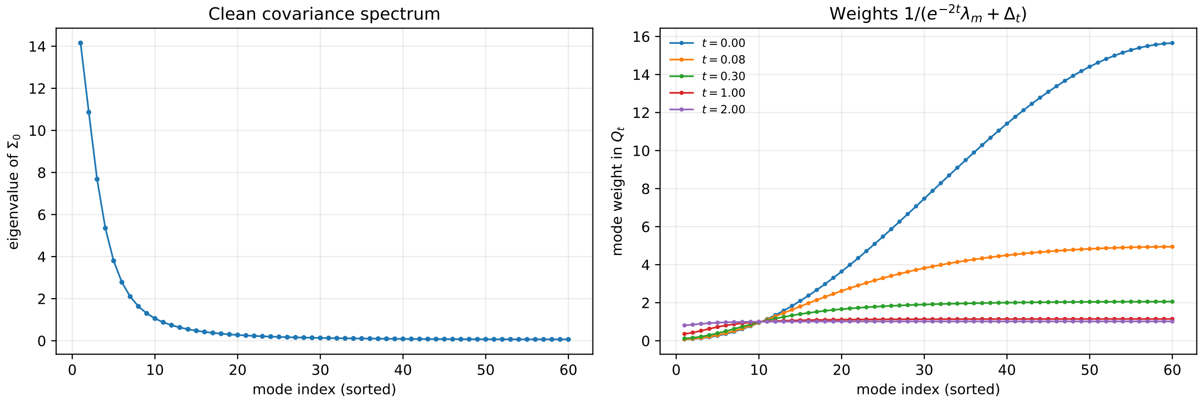

Figure 5: Spectral picture. Left: clean covariance eigenvalues. Right: mode weights in the noisy precision. Diffusion changes the weights without changing the eigenvectors; this breaks the exact cancellation that made Q_0 tridiagonal.

The loss of Markovianity is therefore not mysterious. The clean covariance is dense, and matrix inversion is nonlocal. The exact tridiagonal zeros of Q_0 rely on a very specific cancellation. Diffusion perturbs the spectral weights, breaks that cancellation, and the zeros fill in. The resulting matrix remains structured—dense, but exponentially decaying—until isotropic noise eventually removes all coupling altogether.